



OBJECT ORIENTED SOFTWARE ENGINEERING

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Scenario Based Model

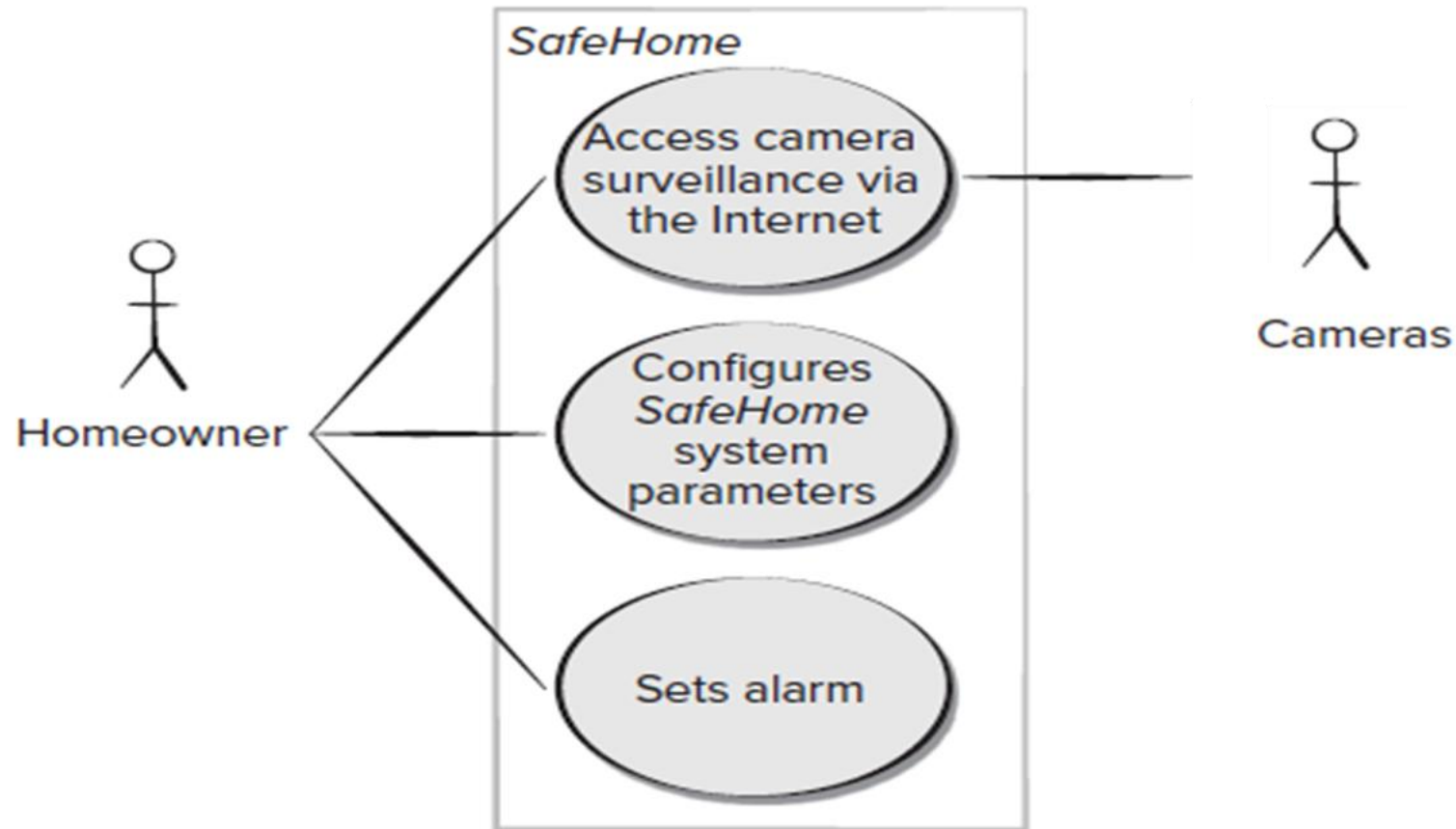
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- **Inception and elicitation**—provides the information needed to begin writing use cases.
- **Requirements gathering meetings, QFD, and other requirements engineering mechanisms** are used to
 - identify stakeholders
 - define the scope of the problem
 - specify overall operational goals
 - establish priorities
 - outline all known functional requirements, and
 - describe the things (objects) that will be manipulated by the system.
- To begin developing a set of use cases, **list the functions or activities performed by a specific actor.**

- As further conversations with the stakeholders progress, the requirements gathering team develops use cases for each of the functions noted.
- In general, use cases are written first in an **informal** narrative fashion (**preliminary use case diagrams**).
- If more formality is required, the same use case is rewritten using a structured format similar to the one proposed.

- Preliminary use case diagram for the **SafeHome system**



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Developing Use Cases

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- A use case is a **methodology** used in system analysis to **identify**, **clarify**, and **organize** system requirements.
- The use case is made up of a **set of possible *sequences of interactions between systems and users*** in a particular environment and related to a particular goal

- Use case tells a **stylized story** about **how an end user interacts with the system** under a specific set of circumstances.
- Regardless of its form, a use case **depicts the software** or system from the **end user's point of view**.

- **Step One** – Define the set of actors that will be involved in the story
 - **Actors** are **people, devices, or other systems** that use the system or product **within the context of the function** and behavior that is to be described
 - Actors are those that **communicate with the system** or product **directly** and that **are external** to the system itself
- **Step Two** – Develop use cases, where each one answers a set of questions

- Components of Use Case diagram

- Use case:

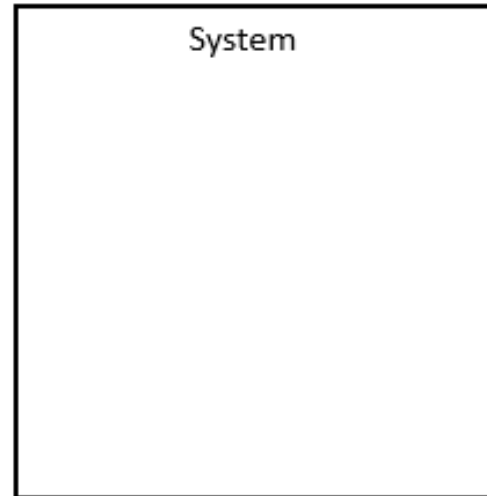


- Subsystem:

- Actor:



- Association:



Actor:- a stick figure labelled with the name of the actor.

Eg. Administrator, Receptionist, verification system etc.

- An actor is a direct external user of a system.

Each actor represents those objects that behave in a particular way towards the system.

Examples:

1.Travel agency system,

Actors:-Traveler, Agent and Airline.

2. Vending machine

Actors:-Customer, Repair technician.

An actor can be **person, device and other system**, anything that interacts directly with the system.

An object can be bound to multiple actors if it has different facts to its behavior.

Eg:- Maya, Vinod, Latha may be customers of a vending machine, Vinod may also be the repair technician interacting with the machine.

The **actor** is directly connected to the system.

Eg.. The **dispatcher** of the **repair technician** from a **service bureau** **is not an actor** of a vending machine, only the repair technician interacts directly with the machine and he is an actor.

Use Cases:

A use case is a coherent piece of functionality that a system can provide by interacting with actors.

Eg:

- A customer actor can **buy a beverage** from a vending machine.
- A repair technician can **perform scheduled maintenance** on a vending machine.

- Each use case involves one or more actors as well as the system itself.

Eg.: - In a telephone system, the use case **make a call** involves two actors, a **caller** and a **receiver**

- An use case involves a **sequence of messages** among the system and its actors.
This can be repeated several times.

Eg.: - In the buy a beverages use case, the customer first inserts a coin and the vending machine displays the amount deposited.

Define a **mainline behavior sequence first**, then optional subsequences, repetitions and other variations.

Eg:-A customer can deposit a variable number of coins in the “buy a beverage” use case, depending on the money inserted and the item selected the machine may or may not return change.

Error condition are also part of a use case.

Eg:- If a customer selects a beverage whose supply is exhausted, the vending machine displays a warning menu

Use Case Diagram:

A system involves a **set of use cases** and **a set of actors**.

Set of use cases:- shows the complete functionality of the system at some level of detail.

Set of actors:- represents the complete set of objects that the system can serve.

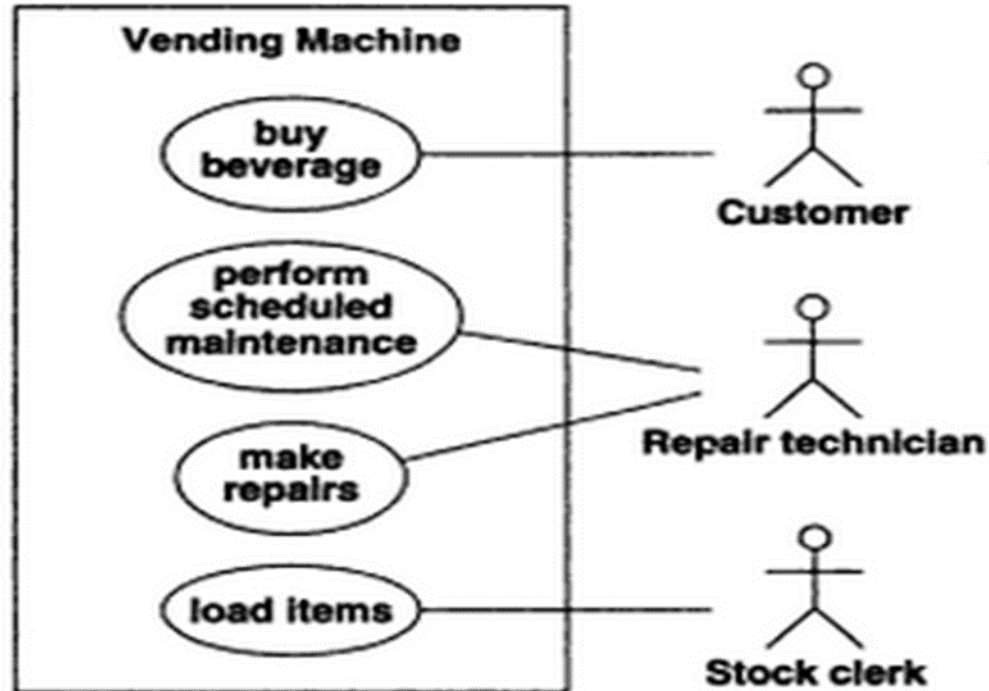


Figure 7.3 Use case diagram for a vending machine. A system involves a set of use cases and a set of actors.

- An use case involves **a sequence of messages** among the system and its actors. This can be repeated several times.

Consider for example **Hotel room booking system**

Actors: Customer, Receptionist, Administrator

Use Cases:

- User login, check room availability, book room, cancel booking, make payment, ask for room booking, check booking, check-in time, check-out time, Manage users, login, manage database, admin manages all the functionalities of the application

Hotel Room booking System

- **Actors:** Customer, Receptionist, Administrator

- **Use Cases:**

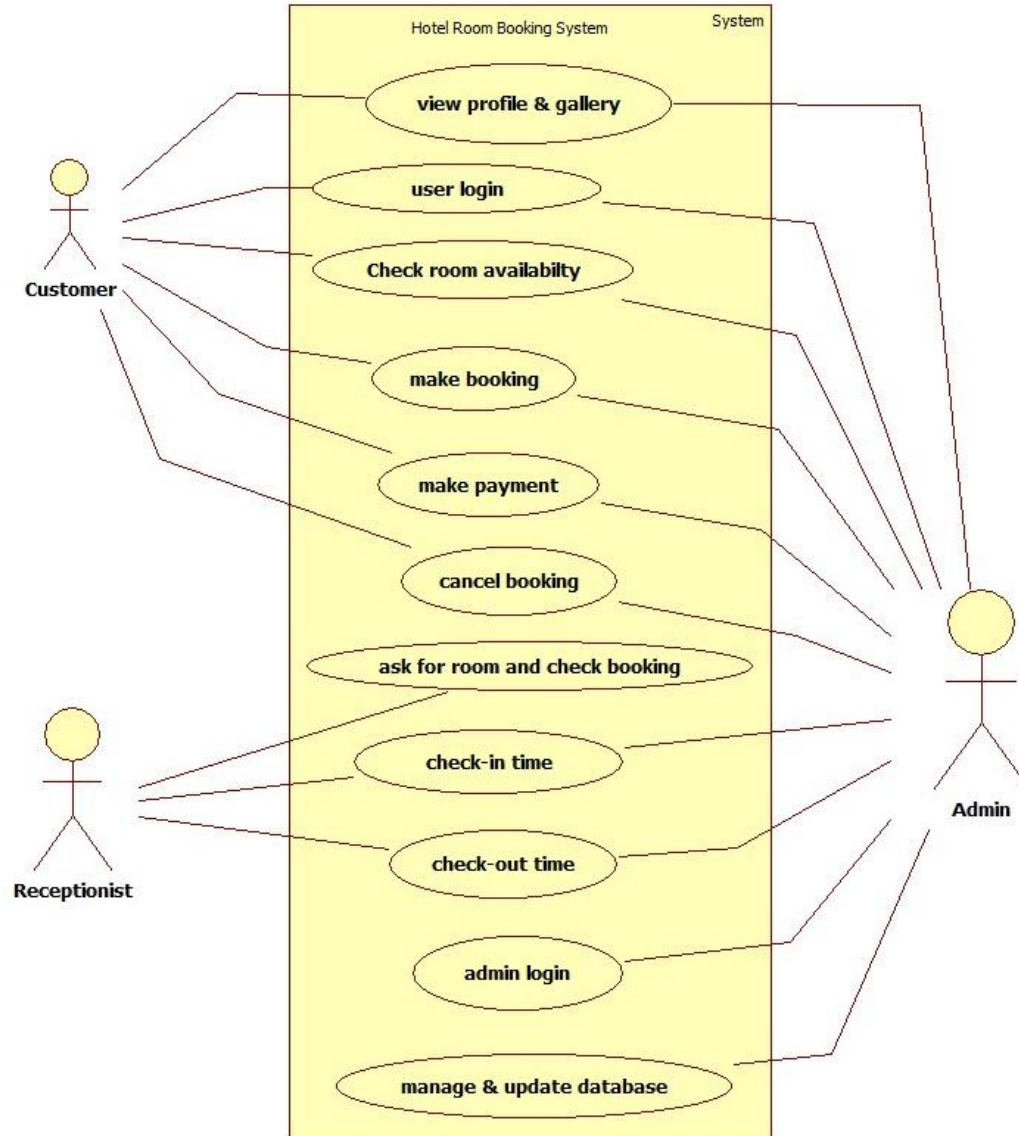
Customer's functionalities are → User login, check room availability, book room, cancel booking, make payment.

Receptionist → ask for room booking, check booking, check-in time, check-out time

Admin → Manage users, login, manage database, admin manages all the functionalities of the application

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Developing Use cases...





THANK YOU

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